Project 6: Patterns in riparian plant community water use

Overview: You will analyze vegetation water use data (evapotranspiration [ET] estimates) that were extracted from the scientific literature. These data were compiled to evaluate broad patterns of vegetation water use across dryland river systems in North America, with the goal of supporting water and land management decisions during water crises (e.g., during droughts or under excess water demand, Palmquist et al., 2026). For this exercise, we will build a hierarchical linear regression model that allows seven vegetation types to respond differently to climate predictors and includes a 2-way interaction between the two climate covariates. The full model description and code are available in Palmquist et al. (2026) and the full data set is published in DiMartini and Palmquist (2025).

The data: The associated data file contains estimates of ET rates in units of mm/year (Annual_ET) along with other variables. For this exercise, we will also use the columns MAT = mean annual temperature (ºC), MAP = mean annual precipitation (mm), and VegID = vegetation type category. Notice that Annual_ET, MAT, and MAP are continuous variables and VegID is a categorical variable represented by integers.

The model: For the likelihood, assume that the log-transformed water use, Y = log(Annual_ET), is normally distributed with a constant variance (or precision, τ) such that for record *i*:

$$y_i \sim Normal(\mu_i, \tau)$$

Specify a simple linear regression model whereby the mean, μ, is defined as a function of MAT, MAP, and their two-way interaction, and allow the regression coefficients (β's) to vary by vegetation type:

$$\mu_i = \beta_{0,v(i)} + \beta_{1,v(i)} MAT_i + \beta_{2,v(i)} MAP_i + \beta_{3,v(i)} MAT_i \cdot MAP_i$$

That is, the β terms, including the intercept, are indexed by $v(i)$, which denotes vegetation community type v associated with observation i (v = 1, 2, 3, …, 7). This model allows for the water use of different vegetation types to respond differently to climate.

Complete the model by specifying priors. We will assign conjugate, hierarchical priors to the vegetation-specific coefficients, such that for coefficient k (k = 0, 1, …, 3) and vegetation type v (v = 1, 2, …, 7):

$$\beta_{k,v} \sim Normal\left(\mu_{\beta_k}, \tau_{\beta_k}\right)$$

Finally, assigned relatively non-informative, conjugate priors to the root nodes for each μ_β, τ_β, and to the residual precision term (τ). Note that the μ_β terms describe the overall ("population-level") climate effects, across all vegetation types, and the τ_β terms describe variability among the vegetation-type-specific climate effects. Thus, for each coefficient k:

$$\mu_\beta \sim Normal(0, 0.0001)$$

$$\tau_\beta \sim Gamma(0.01, 0.01)$$

Code and run the above model via R and Jags for 3 parallel MCMC chains for 10,000 iterations per chain. You can try letting Jag simulate initials for the parameters, or you can attempt to provide initials for the root node parameters. Evaluate MCMC behavior (mixing and convergence) and produce posterior estimates of the model parameters. Do the different vegetation types respond differently to climate?

References Cited

DiMartini, C., and Palmquist, E.C., 2025, Riparian plant and ecosystem water use data in North American drylands between 1997 and 2020 extracted from published studies: U.S. Geological Survey data release, https://doi.org/10.5066/P13BKP5E.

Palmquist, E.C., Nagler, P., Ogle, K., DiMartini, C., Kennedy, J.R., and Sankey, J.B., 2026, JAMES BUTTLE REVIEW: A Synthesis of Riparian Plant Water Use Over Two Decades in North American Drylands: Hydrological Processes, v. 40, no. 2, p. e70408, https://onlinelibrary.wiley.com/doi/abs/10.1002/hyp.70408.